

## TABLE OF CONTENTS

|                                                  |           |
|--------------------------------------------------|-----------|
| <b>TABLE OF CONTENTS</b> .....                   | <b>1</b>  |
| <b>1- REPLACEMENT PROCEDURE</b> .....            | <b>2</b>  |
| 1-1 Tools and Instruments Required .....         | 2         |
| 1-2 Safety .....                                 | 2         |
| 1-3 Preliminary Set-Up Procedure .....           | 4         |
| 1-4 SGA Power Supply Replacement Procedure ..... | 5         |
| 1-5 SGA-PS Module Replacement .....              | 15        |
| 1-6 Functional Tests Required.....               | 17        |
| <b>REVISION HISTORY</b> .....                    | <b>18</b> |

## ACGD SGA POWER SUPPLY REPLACEMENT

### 1- REPLACEMENT PROCEDURE

#### 1-1 Tools and Instruments Required

| Item | Description                                    | Part Number               |
|------|------------------------------------------------|---------------------------|
| 1    | SGA Power Supply                               | 2334862-2                 |
| 2    | Phillips-head Screwdriver                      |                           |
| 3    | Straight-edge Screwdriver                      |                           |
| 4    | Universal Lift Hoist Kit with ACGD Attachments | 46-317266G6, G7, G8 or G9 |
| 5    | Digital Multimeter with High-Voltage Probes    |                           |

#### 1-2 Safety

This procedure requires two (2) FEs.



**FATAL ELECTRIC SHOCK HAZARD! TO PREVENT FATAL ELECTRIC SHOCK, DISCONNECT POWER FROM THE PDU BEFORE PERFORMING THE FOLLOWING PROCEDURES. PERFORM LOCKOUT / TAGOUT PROCEDURE.**



**FATAL ELECTRIC SHOCK HAZARD! THERE IS POTENTIAL 800 VOLTS ON CAPACITORS. WAIT AT LEAST 5 MINUTES AFTER TURNING OFF POWER TO POWER SUPPLY BEFORE SERVICING.**

**WARNING!**

**POSSIBLE PERSONAL INJURY! HEAVY OBJECT – THE POWER SUPPLY WITH CASE WEIGHS APPROXIMATELY 240 POUNDS (109 KG). ASSISTANCE TO LIFT THIS PRODUCT [MECHANICAL (DOLLIES) OR ADDITIONAL PERSONNEL] IS REQUIRED.**

**WARNING!**

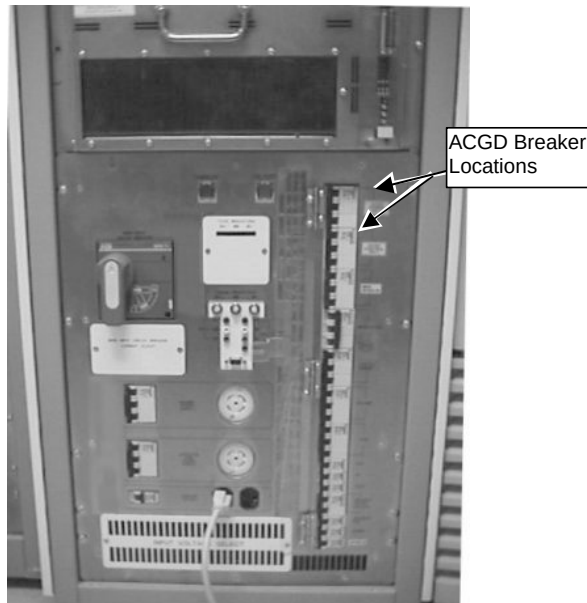
**POSSIBLE PERSONAL INJURY! HEAVY OBJECT - THE HOIST KIT WEIGHS 73 LBS (33 KG). ASSISTANCE TO LIFT THIS PRODUCT [MECHANICAL (DOLLIES) OR ADDITIONAL PERSONNEL] IS REQUIRED. THE HOIST KIT IS REQUIRED TO REMOVE / REPLACE THE MODULE FROM THE CABINET.**



**CABINET TIP HAZARD - POSSIBLE PERSONAL INJURY! CONFIRM THAT THE ALUMINUM ANTI-TIP LEGS STORED IN THE REAR OF THE CABINET ARE INSTALLED ON EITHER SIDE OF THE BASE OF THE CABINET BEFORE ATTEMPTING TO REMOVE AN SGA-POWER SUPPLY.**

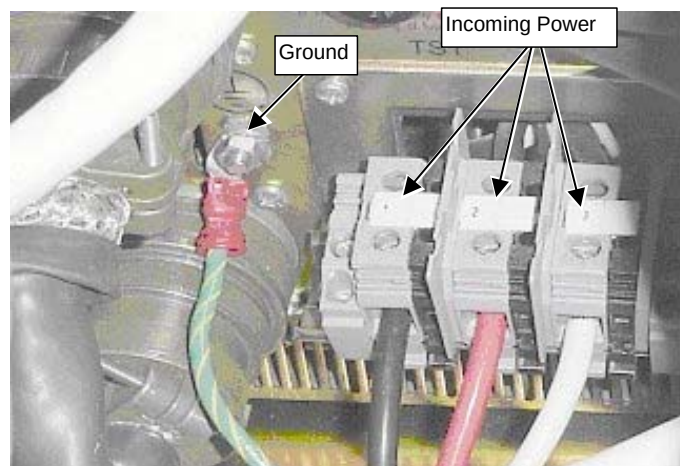
### 1-3 Preliminary Set-Up Procedure

1. Power down the ACGD Cabinet by turning off the ACGD 208V Control Power and the ACGD 420 V Power on the Front Panel of the PDU. (See Illustration 1-1.)



**ACGD CABINET - PDU (FRONT COVER REMOVED)**  
ILLUSTRATION 1-1

2. Lock Out and Tag Out the breaker panel.
3. Verify that power was removed from the cabinet.
  - a) Take a Digital Multimeter and set it to its highest AC voltage range.
  - b) On the terminal block in the rear of the cabinet of the SGA-PS, measure between the ground connection and the incoming power connection at the black wire. (See Illustration 1-2.)

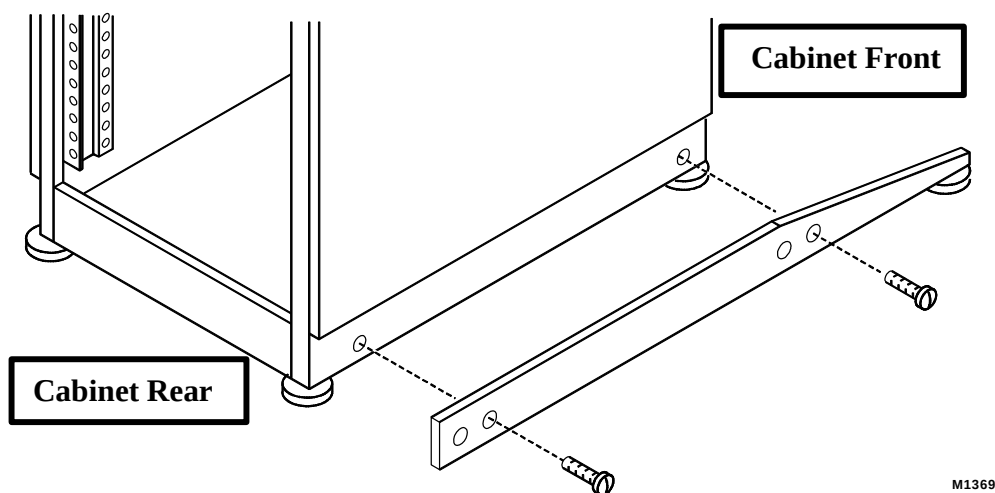


**TERMINAL STRIP**  
ILLUSTRATION 1-2

- c) Repeat this measurement between the ground connection and the red wire, and between the ground connection and the white wire. A measurement of 0 VAC indicates that the 420V incoming power was successfully turned off.

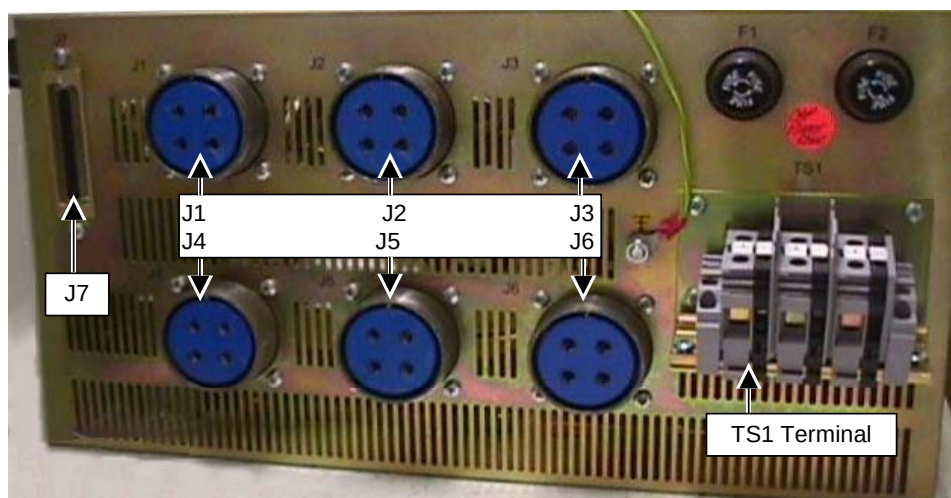
#### 1-4 SGA Power Supply Replacement Procedure

1. Using the screws included with the anti-tip legs, secure the aluminum anti-tip legs to the threaded holes on each side of the base of the ACGD Cabinet. (See Illustration 1-3.)



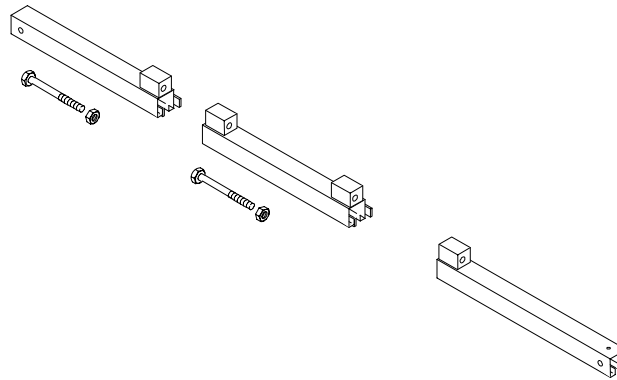
INSTALLING ANTI-TIP LEGS ON SIDES OF CABINET  
ILLUSTRATION 1-3

2. Remove the four pin connectors (J1, J2, J3, J4, J5 and J6) from the rear of the SGA Power Supply Module. (See Illustration 1-4.)
3. Remove the data cable (J7) from the rear of the SGA Power Supply Module. (See Illustration 1-4.)
4. Disconnect the TS1 terminal. (See Illustration 1-4.)



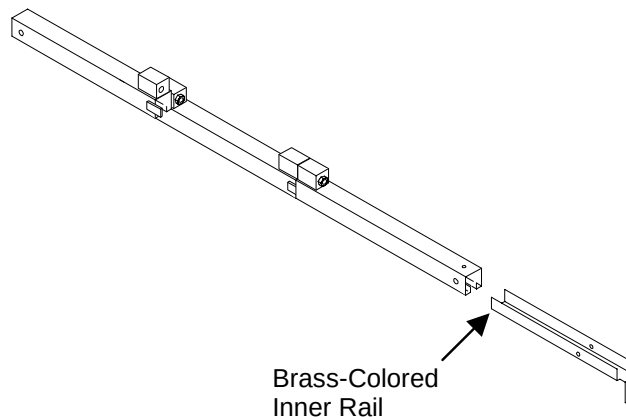
SGA POWER SUPPLY - REAR CONNECTIONS  
ILLUSTRATION 1-4

Locate the three (3) large Steel Rail sections in the Universal Hoist Kit and place them on a flat surface for assembly. (See Illustration 1-5.)



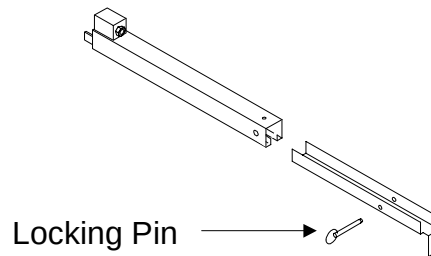
**STEEL RAIL SECTIONS, HEX BOLTS, AND HEX NUTS**  
ILLUSTRATION 1-5

6. Arrange the rail sections in the proper order (rear, center, then front) and fasten them together with 4.75 in. x 1/2 in. caps crews and hex nuts.
7. Use the crescent wrenches to tighten the cap screws.
8. Locate the brass-colored Inner Rail, and insert it into the steel rails as shown in Illustration 1-6.



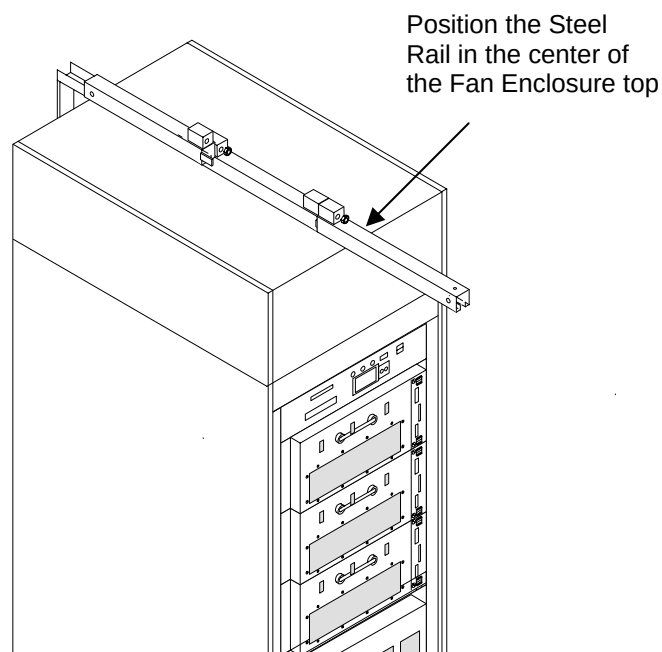
**ASSEMBLED STEEL RAILS AND BRASS-COLORED INNER RAIL**  
ILLUSTRATION 1-6

9. Insert the brass-colored Inner Rail into the open end of the rear rail section.
10. Locate one of two (2) locking pins in the lift kit, and insert one locking pin into the hole near the open end of the rear rail section. This locking pin secures the brass-colored Inner Rail to the Outer Steel Rail. (See Illustration 1-7.)



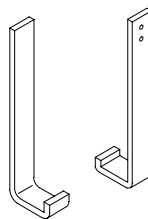
**REAR STEEL RAIL, LOCKING PIN, AND BRASS-COLORED INNER RAIL**  
ILLUSTRATION 1-7

11. Place the Steel Rail assembly on top of the Fan Enclosure at the top of the ACGD Cabinet, and center the Rail Assembly on the top of the enclosure. (See Illustration 1-8.)



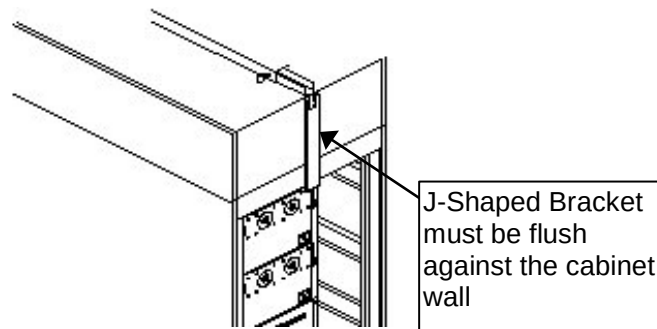
**POSITIONING OF STEEL RAIL ASSEMBLY**  
ILLUSTRATION 1-8

12. Locate the brass-colored J-shaped bracket in the Lift Kit. (See Illustration 1-9.)



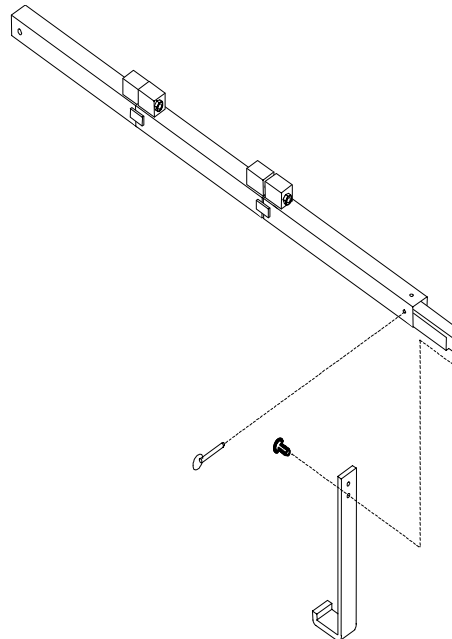
**J-SHAPED BRACKET**  
ILLUSTRATION 1-9

13. Open the rear door of the ACGD Cabinet, and observe how the J-shaped Bracket fits into the rear doorframe. (See Illustration 1-10.)



**POSITIONING OF J-SHAPED BRACKET**  
ILLUSTRATION 1-10

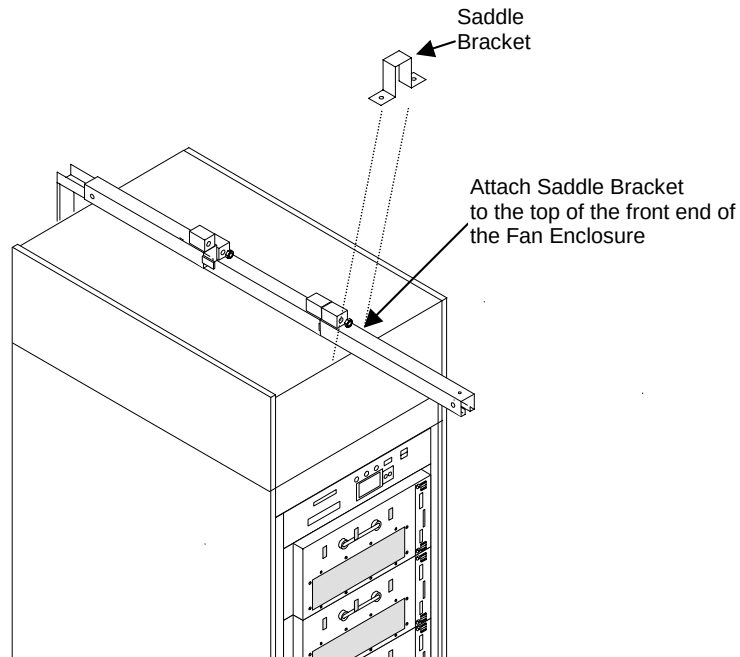
14. Slide the Rail Assembly toward the front of the ACGD Cabinet, maintaining its centered position.
15. Attach the J-shaped Bracket to the brass-colored Inner Rail so that the J-shaped Bracket is flush against the rear panel of the Fan Enclosure. Use the flat Phillips-head screws to secure the J-shaped Bracket to the brass-colored Inner Rail. (See Illustration 1-11.)



**CONNECTION OF J-SHAPED BRACKET TO STEEL RAIL ASSEMBLY**  
ILLUSTRATION 1-11

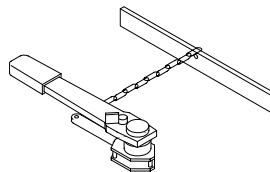


16. Once the J-Shaped Bracket is secured to the cabinet and the Steel Rail, attach the Saddle Bracket to the top of the Fan Enclosure. The screw holes are located at the front end of the Fan Enclosure. The Saddle Bracket prevents the Steel Rail from sliding from center position. (See Illustration 1-12.)



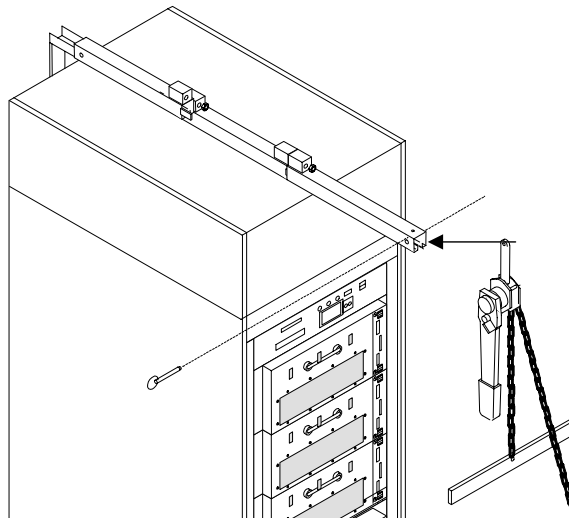
**POSITIONING AND ATTACHMENT OF THE SADDLE BRACKET**  
ILLUSTRATION 1-12

17. Locate the Hoist Assembly in the Lift Kit. (See Illustration 1-13.)



**HOIST ASSEMBLY**  
ILLUSTRATION 1-13

18. Insert the Hoist Assembly into the open end of the front rail, and insert a locking pin into the holes near the front edge of the front rail. This locking pin prevents the Hoist Assembly from sliding out of the rails. (See Illustration 1-14.)



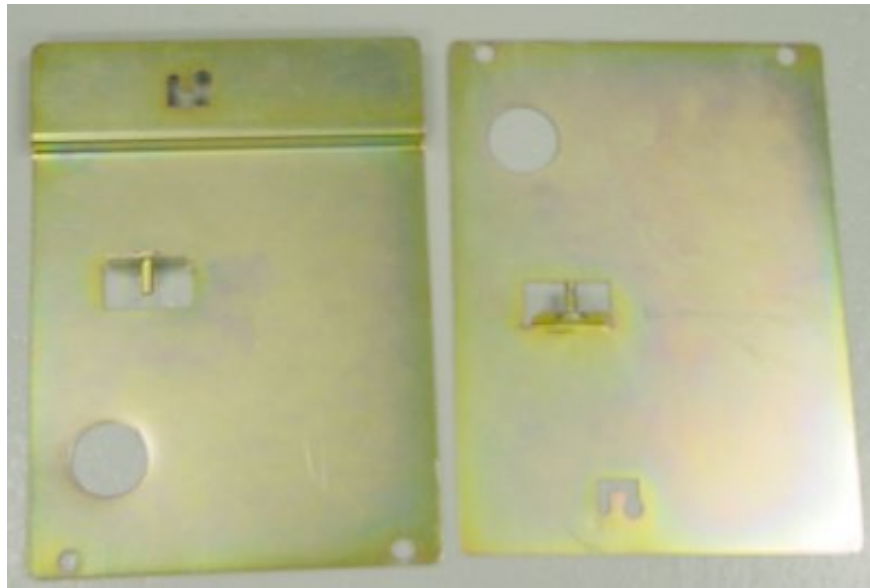
**PLACEMENT OF HOIST ASSEMBLY INTO FRONT END OF STEEL RAIL ASSEMBLY**  
ILLUSTRATION 1-14

19. Remove the four (4) Phillips-head screws and washers from the SGA Power Supply (SGA-PS) front panel that secures the SGA-PS to the ACGD Cabinet. (See Illustration 1-15.)



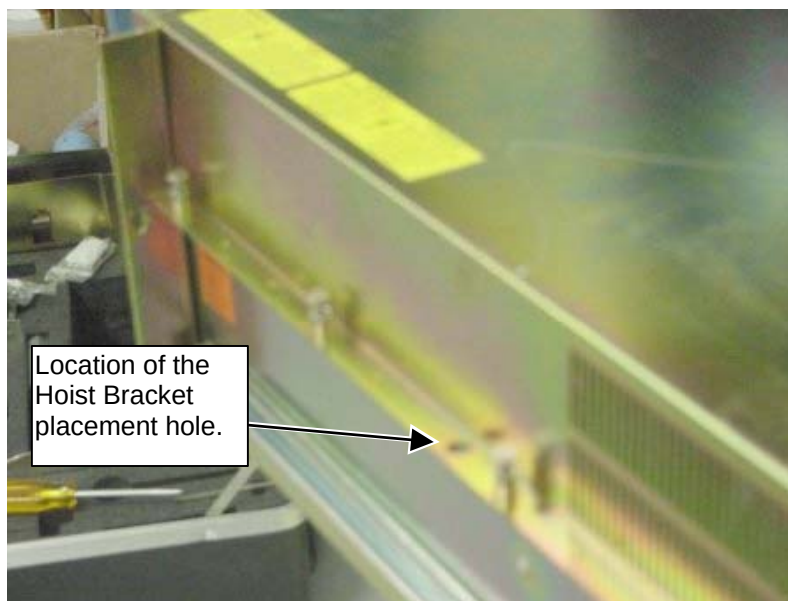
**SCREW LOCATION ON SGA-PS FRONT PANEL**  
ILLUSTRATION 1-15

20. Pull out the SGA-PS Module until its slide rails latch.
21. Remove the two SGA-PS brackets from the Universal Hoist Kit. (See Illustration 1-16.)



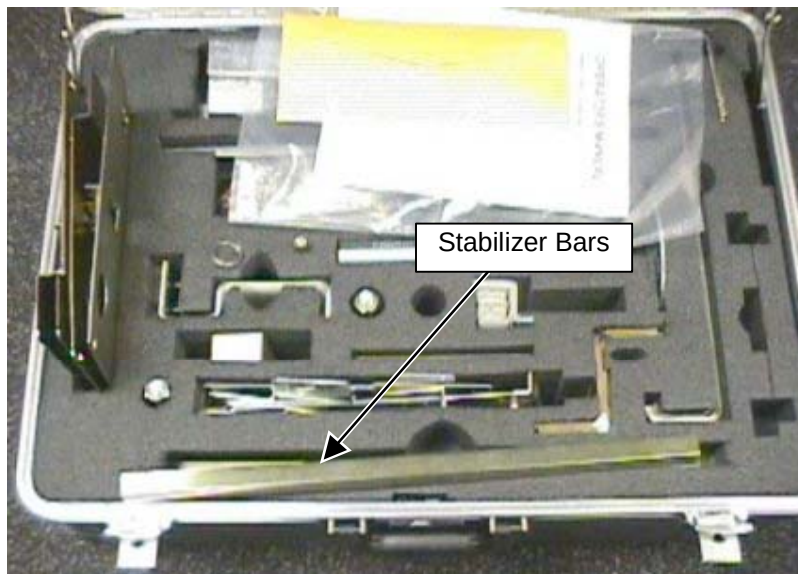
**ACGD SGA-PS HOIST BRACKETS**  
ILLUSTRATION 1-16

22. Brackets are labeled left and right. Place the appropriate bracket in the hole that is on the lip of sheet metal on each side of the SGA-PS. (See Illustration 1-17.)



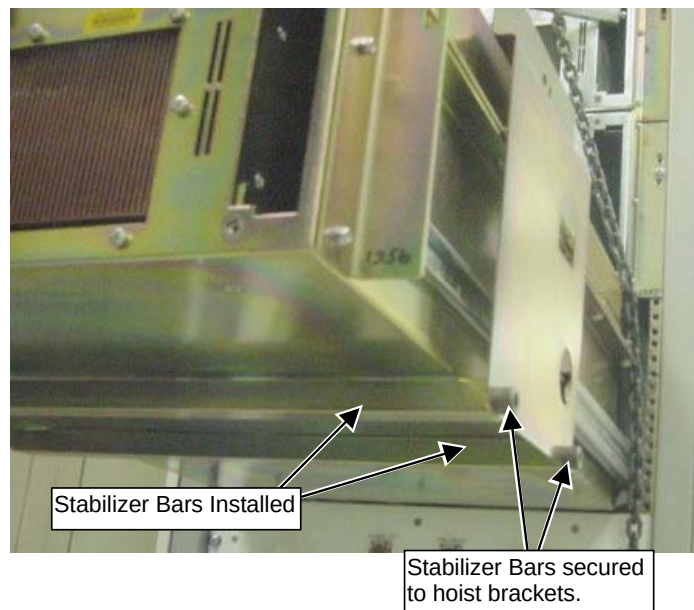
**LOCATION OF HOIST BRACKET PLACEMENT HOLE**  
ILLUSTRATION 1-17

23. Locate the stabilizer bars in the Hoist Kit. (See Illustration 1-18.)



**LOCATION OF STABILIZER BARS IN THE HOIST KIT**  
ILLUSTRATION 1-18

24. Using four screws from the hoist kit, attach the stabilizer brackets to the bottom of both hoist brackets. (See Illustration 1-19.)



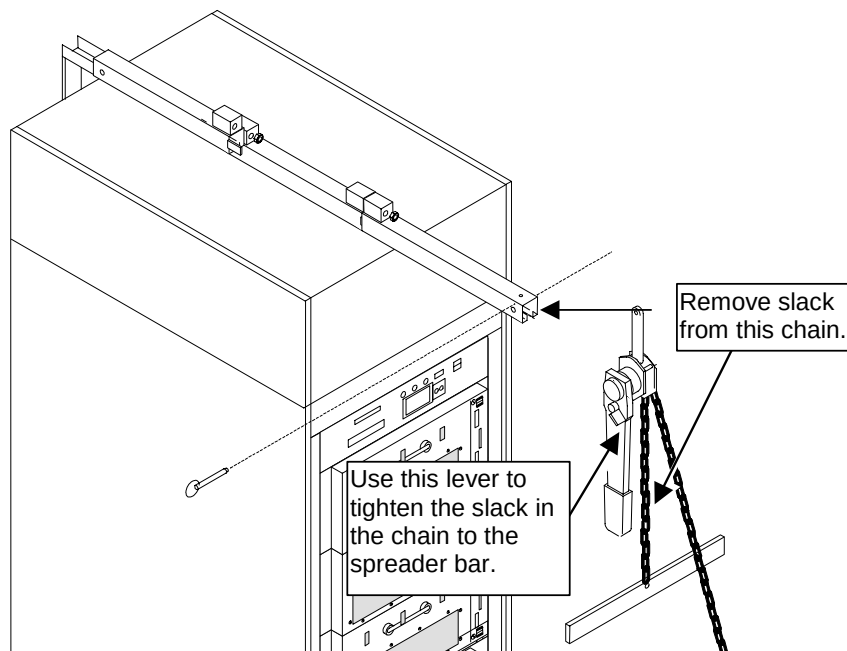
**ATTACHING STABILIZER BARS TO HOIST BRACKETS**  
ILLUSTRATION 1-19

25. Attach the spreader bar of the hoist assembly to the SGA Hoist Brackets. This transfers the weight of the SGAs to the hoist. (See Illustration 1-20.)



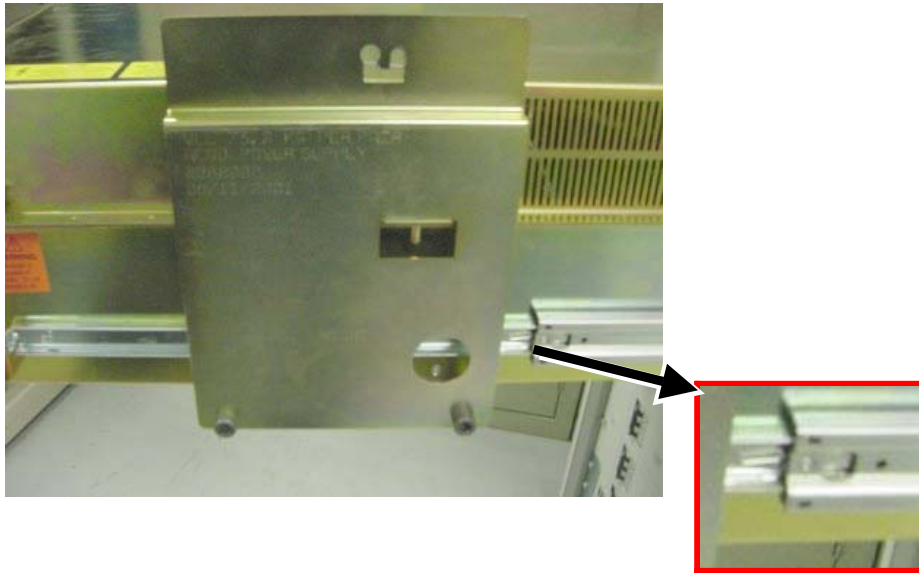
**ATTACHING THE SPREADER BAR TO THE HOIST BRACKETS**  
ILLUSTRATION 1-20

26. On the Hoist Assembly, tighten the chain to the spreader bar to remove slack and support the weight of the SGA module. (See Illustration 1-21.)



**TRANSFERRING WEIGHT TO HOIST ASSMEBLY**  
ILLUSTRATION 1-21

27. Place the rail lever in the UP position on both sides of the SGA-PS. (See Illustration 1-22 for location of lever.)



**RAIL LEVER LOCATION**  
ILLUSTRATION 1-22

28. Ensure that the entire weight of the SGA-PS is supported by the Universal Hoist.
29. Slide the module out, and remove it from the rails that are attached to the cabinet.
30. Turn the module to the side and lower the module. (See Illustration 1-23.)



**EXAMPLE OF A MODULE TURNED TO SIDE**  
ILLUSTRATION 1-23

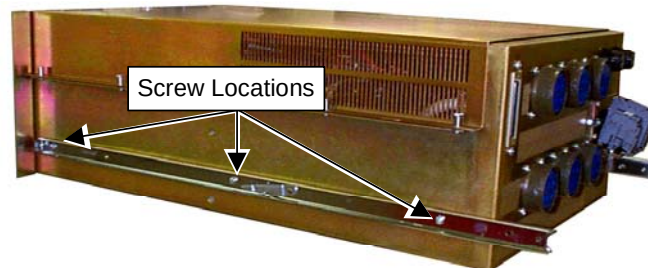
31. Remove the spreader bar from the hoist brackets.



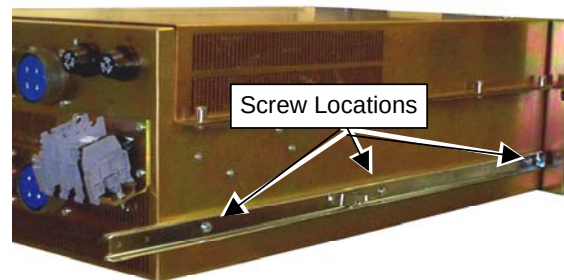
32. Remove the hoist brackets from the failed module.

### 1-5 SGA-PS Module Replacement

1. Remove the rails from the failed module and attach to the replacement module. (See Illustration 1-24.)



Right Side

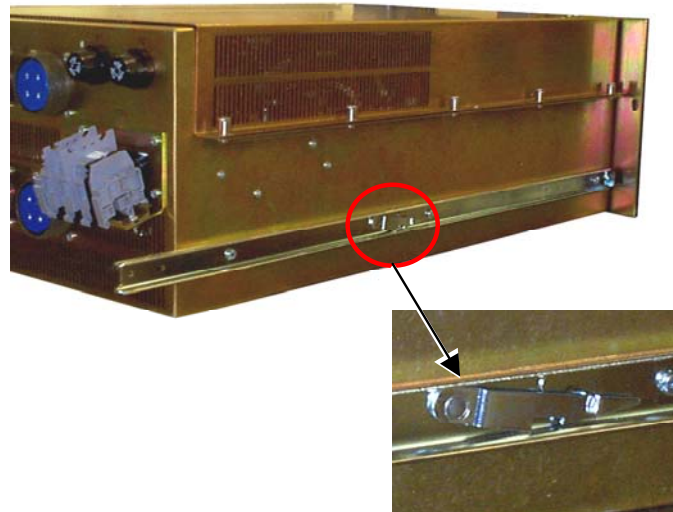


Left Side

**ATTACHING RAILS TO SGA-PS**  
ILLUSTRATION 1-24

2. Attach the hoist brackets to the replacement module.
3. Attach the spreader bar to the hoist brackets.
4. Fully extend the slide rail in the cabinet for the axis that is being replaced.
5. Raise the replacement SGA using the hoist assembly, and move module in position to place rails into the slide mechanism on the cabinet.
6. Install the SGA-PS into the cabinet by inserting the module into the mating half of the rail assemblies.

7. Place the slide-rail lever in the DOWN Position. (See Illustration 1-25.)



**RAIL LEVER LOCATION**  
ILLUSTRATION 1-25

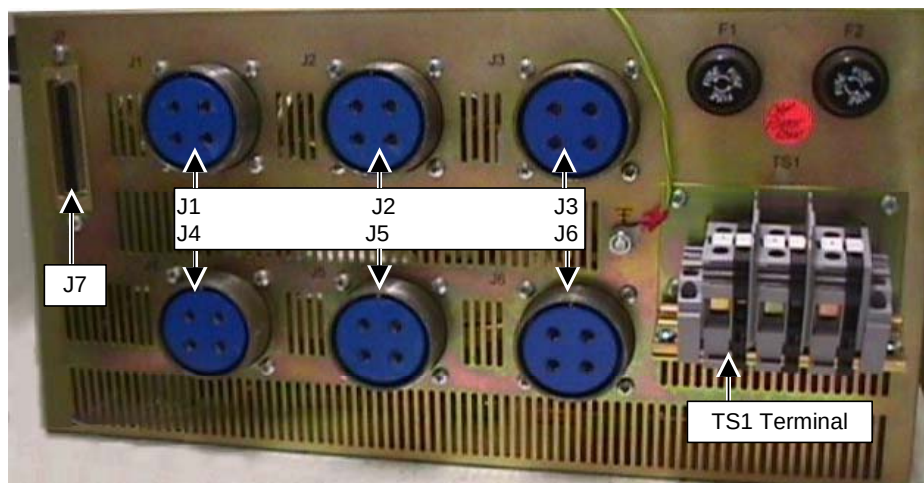
2. Push the SGA-PS assembly all the way into the cabinet.
3. Remove the spreader bar from the hoist brackets.
4. Remove the hoist brackets.
5. Secure the assembly to the front panel by replacing the four (4) Phillips-head screws and washers from the SGA-PS front panel. (See Illustration 1-26.)



**SCREW LOCATION ON SGA-PS FRONT PANEL**  
ILLUSTRATION 1-26



12. Replace the four pin connectors (J1, J2, J3, J4, J5 and J6) at the rear of the SGA-PS Module. (See Illustration 1-27.)
13. Replace the data cable (J7) at the rear of the SGA-PS Module. (See Illustration 1-27.)
14. Connect the TS1 terminal. (See Illustration 1-27.)



**SGA POWER SUPPLY - REAR CONNECTIONS**  
ILLUSTRATION 1-27

15. Remove the Universal Hoist from the cabinet and return to the kit.
16. Remove the power Lock Out / Tag Out to the ACGD Cabinet and restore power.

### 1-6 Functional Tests Required

Power up the cabinet and set up for a scan to verify gradient driver comes up to Ready.

## REVISION HISTORY

| REV | DATE              | AUTHOR     | PRIMARY REASONS FOR CHANGE                                                                                                         |
|-----|-------------------|------------|------------------------------------------------------------------------------------------------------------------------------------|
| A   | August 16, 2000   | K. Keshena | Preliminary version.                                                                                                               |
| 0   | Oct. 22, 2000     | K. Keshena | Initial Release.                                                                                                                   |
| 1   | July 13, 2001     | K. Keshena | Updated procedure to show the SGA-PS module and the removal/replacement for the sliding rails.                                     |
| 2   | June 4, 2003      | K. Keshena | Added steps to use the Universal Hoist Kit to resolve FPR 1020130, SPR ID, MRIge73349.                                             |
| 3   | October 10, 2007  | K. Keshena | Updated the part numbers for the SGA Power Supply per PQR 13082817. Per PSR 13105722, added the Warning message for the hoist kit. |
| 4   | February 29, 2008 | J. Kunze   | Added a Danger statement in Section 1-2 and Warning statements to Section 1-2 and 1-3.                                             |
| 5   | May 5, 2008       | P. Burbach | Per PSR 13106480, changed Section 1-2 to Safety. Moved Warning statements from old Sections 1-2 and 1-3. Renumbered sections.      |